THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · HARD LEVEL UNIT TEST SET 3

Chapters 5–8: **Measurement of Length and Motion, Materials Around Us, Temperature and its Measurement, A Journey through States of Water** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A clinical thermometer reads 36.8 °C and 37.0 °C in two successive tests. This suggests: 1

(a) hypothermia **(b)** error

(c) normal temperature with small variation **(d)** fever

2. A tank of 1000 L loses 2 L per day by evaporation. After 30 days, the water lost is: 1

(a) 30 L **(b)** 60 L **(c)** 600 L **(d)** 2 L

3. Which of these is the best reason for taking repeated measurements? 1

(a) to reduce random errors **(b)** to make the object longer

(c) to change units **(d)** to waste time

4. A bottle cap made of plastic is replaced by a metal cap. Which problem might arise? 1

(a) it may rust or conduct heat **(b)** it dissolves

(c) it becomes transparent **(d)** it floats

5. Which of the following devices uses an insulator to protect the user? 1

(a) iron rod **(b)** electric iron handle

(c) kettle body **(d)** frying pan base

6. Which of the following changes is reversible? 1

(a) cooking of rice **(b)** rusting **(c)** melting of ice **(d)** burning of wood

7. A thread of length 188 cm just goes round a circular plate 4 times in a spiral winding of 4 turns. The circumference is: 1

(a) 47 cm **(b)** 188 cm **(c)** 94 cm **(d)** 752 cm

8. Which of these shows that solubility depends on the substance and not only on appearance? 1

(a) sugar and chalk powder both look white but behave differently in water

(b) both are white

(c) both are powders

(d) both are solid

9. A student records water temperature in the evening: 20 °C, 20 °C, 20 °C after placing in a room at 20 °C. This shows: 1

(a) heating **(b)** thermal equilibrium

(c) cooling **(d)** boiling

10. If the surface area of a liquid is doubled, the rate of evaporation will: 1

(a) increase **(b)** stay the same **(c)** stop **(d)** decrease

11. Which statement about a ceiling fan switched on is correct? 1

(a) the blades move randomly **(b)** the fan is at rest
(c) each blade moves linearly **(d)** each blade moves in a circular path

12. After removing iron filings, adding water to the mixture and filtering separates: 1

(a) salt on the filter paper **(b)** sand on the filter paper
(c) both on the filter **(d)** nothing

13. Which temperature is not possible on a clinical thermometer scale? 1

(a) 20 °C **(b)** 42 °C **(c)** 36 °C **(d)** 40 °C

14. A tap leaks 1 drop per second; if 20 drops make 1 mL, the leak in one hour is about: 1

(a) 180 mL **(b)** 60 mL **(c)** 3600 mL **(d)** 18 mL

15. Parallax error is least when the scale is: 1

(a) viewed from the side
(b) tilted
(c) placed touching the object and viewed straight
(d) placed far away

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Which has more heat: a bucket of water at 40 °C or a cup at 90 °C? Explain. 2

17. How would water table levels change with excessive pumping? 2

18. Why can a circle's circumference not be measured with a straight ruler directly? 2

19. Why does a metal spoon in a cup of hot tea become hot but a plastic one hardly changes? 2

20. A metal spoon and a wooden spoon at room temperature feel different to touch. Does this mean they have different temperatures? 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Plan a fair test to find which material keeps water hot longest: wool, cotton, aluminium foil. 3

22. Explain with an experiment how to prove that the water which disappears from a pond goes to the air. 3

23. Explain, with two examples each, the difference between rotation and revolution. 3

24. Give three reasons why plastic has replaced many materials in daily life. 3

OR

Explain how a daily temperature chart can help predict the next day's weather (basic idea). 3

25. Describe how an igloo can survive in cold weather and not melt. 3

OR

Why do we say that a train passenger and platform observer disagree about speed? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. A pendulum of length 1 m makes 30 oscillations in 60 s and a pendulum of length 0.25 m makes 30 oscillations in 30 s. Calculate their time periods and compare. 5

OR

A group of students wanted to prepare a separation plan for a mixture of iron filings, sand, salt and water-soluble blue dye. Write the full flow and explain the principle at each step. 5

27. Explain how heat is transferred in a metal spoon placed in boiling soup and why the plastic spoon stays cool. 5

OR

Explain how a solar still can convert sea water to drinking water, naming the changes of state. 5

END OF QUESTION PAPER — Total: 50 Marks

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